

Inductive Bias

Inductive Bias is the "Prior knowledge" or a set of assumptions dumped into a model's architecture, learning algorithm, or optimization process that steer it toward some specific solutions over others, enabling generalization beyond the training data.

Example:

- **Linear regression:** this algorithm assumes that relationship between input and output is linear. and it cannot represent curves at all.
- **K-nearest neighbors:** assumes that points close together in feature space should have similar labels (a smoothness assumption)
- **Convolutional neural networks (CNNs):** biased toward spatial locality and translation invariance which is an assumption that a pattern (like an edge or a cat's ear) mean the same thing regardless of where it appears in an image. This is why CNNs work well on images with less data than a fully-connected network would need
- **Decision trees:** biased toward axis-aligned splits and tend to prefer simpler trees (fewer splits) when regularized
- **Bayesian models** encode inductive bias explicitly through the choice of prior distribution.

Why it is matter

- **No free lunch theorem:** there's no single learning algorithm that performs best across all possible problems. Good performance on a specific type of problem requires an inductive bias that matches the structure of that problem.
- **Bias-variance tradeoff:** strong inductive bias tends to reduce variance (less overfitting, more stable predictions) but can increase bias (risk of underfitting if the assumption is wrong for the given task).
- **Architecture design in deep learning** is largely about choosing the right inductive biases like CNNs for images, RNNs/Transformers for sequences (though transformers have weaker built-in biases and rely more on data + attention to learn structure), graph neural networks for graph-structured data, and so on.